INCH – POUND
MIL-DTL-70788B(AR)
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SUPERSEDING
MIL-A-70788A (AR)
W/AMENDMENT 5
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DETAIL SPECIFICATION

ADAPTER, METAL AND WOOD PALLET

This specification is approved for use by the U.S. Army Armament Center and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations, tests, and packaging of special adapters for wood or metal pallets. The adapters, together with the wood or metal pallet, will be used in unitizing ammunition (See 6.1).

1.2 Classification. The adapters will be of the following types:

Type I	-	Adapter, Pallet, PA113 Container
Type II	-	Adapter, Pallet, PA116/171 Container
Type III	-	Adapter, Pallet, PA117 Container
Type IV	-	Adapter, Pallet, PA120 Container
Type V	-	Adapter, Pallet, PA125 Container
Type VI	-	Adapter, Pallet, PA150 Container
Type VII	-	Adapter, Pallet, PA151 Container
Type VIII	-	Top Assembly – Pallet Adapter PA150 Container For use with Wood Pallet
Type IX	-	Top Assembly – Pallet Adapter PA151 Container for use with Wood Pallet
Type X	-	Top Lift adapter, M643 Ammunition Container
Type XI	-	Adapter, Pallet, PA161, Container
Type XII	-	Adapter, Pallet, PA103A2 Container

Comments, suggestions, or questions on this document should be addressed to: U.S Army Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, NJ 07806-5000 or emailed usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-C-490	-	Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-53072	-	Chemical Agent Resistant Coating (CARC) System Applications Procedures and Quality Control Inspection

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-1168	-	Ammunition Lot Numbering and Data card
MIL-STD-1660	-	Ammunition Unit Loads
MIL-STD-1916	-	Department of Defense Test Method Standard: DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

US ARMY DEFENSE AMMUNITION CENTER DRAWINGS

ACV00053	-	Combination of Adopted Items, Pallet and Pallet Adapter, Unitization of PA125 Container
ACV00054	-	Top Assembly - Pallet Adapter PA125 Container
ACV00055	-	Bottom Assembly - Pallet Adapter PA125 Container

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ACV00056 - Intermediate Assembly - Pallet Adapter PA125 4ontainer
ACV00060 - Lifting Bar
ACV00067 - Combination of Adopted Items, Pallet, and Pallet Adapter, Unitization of PA103E2 Container
ACV00068 - Top Assembly, Pallet Adapter, PA103E2 Container
ACV00069 - Bottom Assembly, Pallet Adapter, PA103E2 Container
ACV00124 - Combination of Adopted Items, Pallet and Pallet Adapter, Unitization of PA120 Container
ACV00125 - Top Assembly - Pallet Adapter, PA120 Container
ACV00126 - Bottom Assembly - Pallet Adapter, PA120 Container
ACV00130 - Lifting Ring
ACV00134 - Combination of Adopted Items, Pallet & Pallet Adapter, Unitization of PA117 Container
ACV00135 - Top Assembly - Pallet Adapter, PA117 Container
ACV00136 - Bottom Assembly - Pallet Adapter, PA117 Container
ACV00155 - Combination of Adopted Items, Pallet and Pallet Adapter, Unitization of PA151 Container
ACV00156 - Combination of Adopted Items, Pallet and Pallet Adapter, Unitization of PA150 Container
ACV00157 - Top Assembly - Pallet Adapter, PA151 Container
ACV00158 - Bottom Assembly - Pallet Adapter, PA151 Container
ACV00159 - Top Assembly - Pallet Adapter, PA150 Container
ACV00160 - Bottom Assembly - Pallet Adapter, PA150 Container
ACV00283 - Combination of Adopted Items, Pallet & Pallet Adapter, Unitization of PA160 Container
ACV00284 - Top Assembly - Pallet Adapter PA160 Container
ACV00285 - Bottom Assembly - Pallet Adapter PA160 Container
ACV00294 - Lifting Bar
ACV00300 - Combination of Adopted Items, Pallet & Pallet Adapter, Unitization of PA161E1 Container
ACV00301 - Top Assembly - Pallet Adapter, PA161E Container
ACV00302 - Bottom Assembly - Pallet Adapter, PA161E Container
ACV00515 - General Welding Procedures
ACV00636 - Top Assembly - Pallet Adapter, PA161 Container without Lugs
ACV00637 - Top Assembly - Pallet Adapter, PA103E2 Container without Lugs
ACV00647 - Top Assembly - Pallet Adapter PA150 Container for use with Wood Pallet
ACV00648 - Top Assembly – Pallet Adapter PA151 Container for use with Wood Pallet
ACV00811 - Combination of Adopted Items, Pallet and Pallet Adapter, Unitization of PA116/PA171 Container
ACV00833 - Combination of Adopted Items for Unitization of M643 Ammunition Container
ACV00834 - Bottom Frame and Pallet - M643 Ammunition Container
ACV00835 - Top Lift Adapter - M643 Ammunition Container
ACV00861 - Combination of Adopted Items, Pallet & Pallet Adapter, Unitization of Excalibur in PA194Containers
ACV00862 - Top Assembly - Pallet Adapter PA194 Container

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ACV00863	-	Pallet Assembly – Special Size 47.00 X 33.57 Sheet Metal
AC200000414	-	Combination of Adopted Items, Pallet & Pallet Adapter, Unitization of PA113 Container
AC200000423	-	Chemical Agent Resistant Coating (CARC) Finishing Requirements for Metal Pallets and/or Pallet Adapters
AC200000461	-	Wood Pallet and Top Adapter Assembly for PA116/PA171 Container
AC200000469	-	Top Assembly - Pallet Adapter PA116/PA171 Container
AC200000470	-	Bottom Assembly - Pallet Adapter PA116/PA171 Container
AC200000501	-	Adapter - Pallet, PA116/PA171 Container
AC200000804	-	Lifting Ring
AC200000805	-	Lifting Ring Retainer
AC200000806	-	Lifting Ring Retainer
AC200000807	-	Top Adapter Assembly - PA116/PA171 Container
AC200000812	-	40 X 44 Adapter Lug Positioning Test Fixture
AC200000827	-	Top Assembly - Pallet Adapter PA113 Container
AC200000828	-	Bottom Assembly - Pallet Adapter PA113 Container

(Copies of these drawings are available via email at mailto:usarmy.pica.devcom-ac.list.drawing-request-help-desk@army.mil)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D4138	Measurement of Dry Film Thickness of Protective Coating System by Destructive Means

(Copies of these documents are available from https://www.astm.org/)

SAE INTERNATIONAL

SAE-AMS-STD-595	Colors used in Government Procurement

(Copies of these documents are available from https://www.sae.org/)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. When specified (see 6.2), a sample of the adapters shall be subjected to first article in accordance with section 4.3, Table II and Table III.

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3.2 Conformance inspection. A sample of the adapters shall be subjected to conformance inspection in accordance with section 4.4.

3.3 Adapters. The adapter assemblies shall comply with all requirements in accordance with Drawings AC200000414, AC200000461, AC200000501, ACV00053, ACV00124, ACV00134, ACV00155, ACV00156, ACV00636, ACV000637, ACV00647 and ACV00648.

3.4 Welding. The welding shall comply with the requirements of ACV00515, Class 1.

3.5 Lifting ring (bar) pull test. Lifting rings (bars) of the adapter top assembly shall withstand a (non-destructive) load of not less than 2,500 pounds for a duration of not less than 30 seconds minimum.

3.6 Painted surface. The painted surfaces of the adapters shall comply with the requirements of Drawing AC200000423. When specified (see 6.2) in the contract, the adapters for PA116/PA171 Container (Drawing AC200000461) is the only adapter which has the option of complying with CARC requirements or with requirements for conventional coating.

3.6.1 Conventional coating. When specified (see 6.2), the painted surface of the adapter shall be in accordance with TT-C-490 and a MIL-STD-171.

3.7 Unit load. The metal pallet adapters, as part of unit loaded pallets, shall conform to section 5.2 satisfactory performance criteria of MIL-STD-1660. The unit load shall remain intact and be capable of continued safe handling and tiering.

3.8 Workmanship.

3.8.1 Metal defects. The adapters shall be smooth, and free from wrinkles, cracks, rough spots, burrs, and sharp edges. Smooth weld spatter is permissible on non-load bearing surfaces provided it is firmly attached.

3.8.2 Paint coating. The finished coat shall cover all required surfaces and be continuous. Paint runs are permissible provided they are no more than 3 inches in length, solid and free from cracks and blisters. Excess thickness is acceptable in areas where multiple passes of the paint gun are required to obtain total coverage. Paint bare spots which expose primer or bare metal are acceptable provided they are not greater than 0.25 inches in diameter and no more than two such paint bare spots on an adapter. Scratches which expose primer or bare metal are permissible provided they are no more than 1.5 inches in length and no more than 10 inches total on an adapter. Bare metal on contact surfaces in areas inaccessible to paint is permissible.

3.8.3 Foreign matter. All parts and assemblies shall be free of bare spots, dirt, grease, rust and foreign material. The cleaning method used shall not be injurious to any of the parts nor shall any of the parts be contaminated by the cleaning agents used.

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4. VERIFICATIONS

TABLE I. Requirement/verification cross reference matrix.

METHOD OF VERIFICATION 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test		CLASSES OF VERIFICATION A – First article inspection B – Conformance inspection						
Section 3 Requirement		Verification Methods				Verification Class		Section 4 Verification Procedures
		1	2	3	4	A	B	
3.1	First article		X	X	X	X		4.3
3.2	Conformance inspection		X	X	X		X	4.4
3.3	Adapters		X	X	X	X	X	4.4.1.1 to 4.4.1.8
3.4	Welding		X	X	X	X	X	4.4.1.2, 4.4.1.4, 4.4.1.6
3.5	Lifting ring (bar) pull test		X	X	X	X	X	4.5.2
3.6	Painted surface		X	X	X	X	X	4.4.1.3, 4.5.3
3.6.1	Conventional coating		X	X	X	X	X	MIL-STD-171
3.7	Unit load test		X	X	X	X		4.5.1
3.8	Workmanship		X	X	X	X	X	4.4.1.1 to 4.4.1.8
3.8.1	Adapters		X	X	X	X	X	4.4.1.1 to 4.4.1.8
3.8.2	Paint coating		X	X	X	X	X	4.5.3
3.8.3	Parts			X	X	X	X	4.4.1.1 to 4.4.1.8

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in 4.5.

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4.3 First article inspection. A sample of the adapters and components shall be subjected to first article verification in accordance with Tables II and III.

4.3.1 First article rejection. If any assembly, component, or test specimen fails to comply with requirements, the first article sample shall be rejected.

TABLE II. First article quantity.

Part description	Quantity
Lifting ring (bar)	6
Lifting Ring Retainer	6
Top assembly (unpainted)	6
Bottom assembly (unpainted)	6
Top assembly (painted)	6
Bottom assembly (painted)	6
Intermediate assembly (unpainted)	6
Intermediate assembly (painted)	6
Adapter assembly	3 1/
Unit Load Test	3 1/

Note 1/ The samples for the adapter assemblies shall be comprised of the individual painted top, bottom and intermediate assemblies submitted along with appropriate pallet base.

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TABLE III – First article tests and inspection.

Examination or Test	Quantity	Requirement Paragraph	Inspection Methods
Lifting ring (bar) (Dwg. ACV00294, AC200000804, ACV00060 or ACV00130, as applicable) Examination for defects	6	3.3	4.4.1.1
Lifting Ring Retainer (Dwg. AC200000805, AC200000806) Examination for Defects	6	3.3	4.4.1.8
Top assembly, unpainted (Dwg. AC200000469, AC200000807, AC200000827, ACV00054, ACV00125, ACV00135, ACV00157, ACV00159, ACV00636, ACV00637, ACV00647 or ACV00648 as applicable) Examination for defects	6	3.3	4.4.1.2
Top Assembly, Painted (Dwg. AC200000469, AC200000807, AC200000827, ACV00054, ACV00125, ACV00135, ACV00157, ACV00159, ACV00636, ACV00637, ACV00647 or ACV00648 as applicable) Examination for defects	6	3.3	4.4.1.3
Bottom assembly, unpainted (Dwg. AC200000470, AC200000828, ACV00055, ACV00126, ACV00136, ACV00158 or ACV00160 as applicable) Examination for defects	6	3.3	4.4.1.4
Bottom assembly, painted (Dwg. AC200000470, AC200000828, ACV00055, ACV00126, ACV00136, ACV00158 and ACV00160 as applicable) Examination for defects	6	3.3	4.4.1.5

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TABLE III – First article tests and inspection - Continued.

Examination or Test	Quantity	Requirement Paragraph	Inspection Methods
Intermediate assembly, unpainted (Dwg. ACV00056) Examination for defects	6	3.3	4.4.1.6
Intermediate assembly, painted (Dwg. ACV00056) Examination for defects	6	3.3	4.4.1.7
Adapter Assembly (Dwg. AC200000414, AC200000461, AC200000501, ACV00053, ACV00067, ACV00124, ACV00134, ACV00155, ACV00156, ACV00300, ACV00647 or ACV00648) Unit load test	3	3.7	4.5.1

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4.4 Conformance inspection.

4.4.1 Conformance verification (Lot Acceptance Tests). Sample components and adapters shall undergo conformance inspection by examination, analysis, demonstration, and tests as specified in accordance with Table IV (4.4.1.1 to 4.4.1.8), and applicable paragraphs within this section. For conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

TABLE IV – Conformance inspection.

4.4.1.1 Lifting ring (bar)1/2/	Conformance Criteria		Requirement Paragraph	Inspection Methods
	Qty	MIL-STD-1916 level		
Minor 201 – Evidence of poor workmanship		III	3.8	Visual
Notes: 1/ Drawings AC200000804, ACV00060, ACV00130, ACV00155, ACV00156, ACV00294 applies. 2/ Next higher assembly Drawings AC200000469, AC200000807, AC200000827, ACV00054, ACV00067, ACV00125, ACV00135, ACV00155, ACV00156, ACV00300, ACV00636, ACV00637 applies.				

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TABLE IV – Conformance inspection - Continued.

4.4.1.2 Top Assembly, Unpainted 1/2/	Conformance Criteria		Requirement Paragraph	Inspection Methods
	Qty	MIL-STD-1916 level		
Major 101 – Lifting ring pull test		V	3.3	4.5.2
Major 102 – True position of stacking lugs		IV	3.3	Gage
Major 103 – Welding missing, incomplete, or incorrect		IV	3.4	Visual
Major 104 – Overall length		III	3.3	Gage
Major 105 – Overall width		III	3.3	Gage
Major 106 – Ring minimum height at rest (horizontal position) 3/		III	3.3	Gage
Major 107 – Material 4/			3.3	Visual
Minor 201 – Evidence of poor workmanship		II	3.8.1/3.8.3	Visual

NOTES:

1/ Drawings AC200000469, AC200000807, AC200000827, ACV00054, ACV00125, ACV00135, ACV00157, ADV00159, ACV00636, ACV00637, ACV00647 or ACV00648 applies.

2/ Next higher assembly Drawings AC200000414, AC200000461, AC200000501, ACV00053, ACV00124, ACV00134, ACV00155, ACV00156, ACV00300, ACV00067.

3/ Major characteristic 106 is not applicable to Drawings ACV0054 and ACV00125. Dimension specified on Drawing AC200000805/806 Sheet 2.

4/ Per lot of material.

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TABLE IV – Conformance inspection - Continued.

4.4.1.3 Top Assembly, Painted 1/2/	Conformance Criteria		Requirement Paragraph	Inspection Methods
	Qty	MIL-STD-1916 level		
Major 101 – Topcoat thickness (total thickness minus primer thickness (see AC200000423 and Note 3))		III	3.6	Gage
Major 102 – Marking missing, incorrect or illegible		III	3.3/3.6	Visual
Major 103 – Paint adhesion test	4		3.6	4.5.3
Major 104 – Primer coat thickness (see AC00000423 and Note 3)		III	3.6	Gage
Major 105 – Ring (at rest) below top surface 4/		III	3.3	Gage/Manual
Major 106 – Ring free to rotate from vertical position to resting (horizontal position) with no force applied 4/		III	3.3	Gage/Manual
Minor 201 – Evidence of poor workmanship		II	3.8.2	Visual

NOTES:

1/ Drawings AC200000469, AC200000807, AC200000827, ACV00054, ACV00125, ACV00135, ACV00157, ACV00159, ACV00636, ACV00637, ACV00647 OR ACV00648 applies

2/ Next higher assembly Drawings AC200000414, AC200000461, AC200000501, ACV00053, ACV00067, ACV00124, ACV00134, ACV0155, ACV00156, or ACV00300 applies

3/ The inspection is only applicable for CARC coating. The primer thickness may be determined from a test panel when representative of the part for CARC topcoat. When destructive testing is done on samples (e.g. ASTM-D4138), these surfaces may be touched up in accordance with 3.6.1 of MIL-DTL-53072

4/ Major characteristics 105 and 106 are not applicable to Dwg. ACV0054 and ACV00125.

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TABLE IV – Conformance inspection - Continued.

4.4.1.4 Bottom Assembly, Unpainted 1/2/3/	Conformance Criteria		Requirement Paragraph	Inspection Methods
	Qty	MIL-STD-1916 level		
Major 101 – True position of stacking lugs		VI	3.3	Gage
Major 102 – Weld missing, incomplete or incorrect		VI	3.4	Visual
Major 103 – Overall length		III	3.3	Gage
Major 104 – Overall width		III	3.3	Gage
Major 105 – Material 4/			3.3	Visual
Minor 201 – Evidence of poor workmanship		II	3.8.1/3.8.3	Visual

NOTES:

1/ Drawings AC200000470, AC200000828, ACV00055, ACV00126, ACV00136, ACV00158 or ACV00160 applies, as applicable.

2/ Next higher assembly Drawings AC200000414, AC200000501, ACV00053, ACV00067, ACV00124, ACV00134, ACV00155, ACV00156 or ACV00300 applies, as applicable.

3/ When this inspection is performed on the top assembly for the Type II or Type III Pallet Adapter, the gage shall be as specified in Dwg AC200000812.

4/ Per lot of material.

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TABLE IV – Conformance inspection - Continued.

4.4.1.5 Bottom Assembly, Painted 1/2/	Conformance Criteria		Requirement Paragraph	Inspection Methods
	Qty	MIL-STD-1916 level		
Major 101 – Primer coat thickness (see AC200000423 and Note 3)		III	3.3	Gage
Major 102 – Marking missing, incorrect or illegible.		III	3.3	Visual
Major 103 – Paint adhesion test	4		3.5	4.5.3
Major 104 – Topcoat thickness (total thickness minus primer thickness (see AC200000423 and Note 3))		III	3.3	Gage
Minor 201 – Evidence of poor workmanship		II	3.8.1/3.8.3	Visual

NOTES:

1/ Drawings AC200000470, AC200000828, ACV00055, ACV00126, ACV00136, ACV00158 or ACV00160 applies, as applicable

2/ Next higher assembly Drawings AC200000414, AC200000501, ACV00053, ACV00124, ACV00134, ACV00155 or ACV00156 applies, as applicable

3/ The inspection is only applicable for CARC coating. The primer thickness may be determined from a test panel when representative of the part for CARC topcoat. When destructive testing is done on samples (e.g. ASTM-D4138), these surfaces may be touched up in accordance with 3.6.1 of MIL-DTL-53072

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TABLE IV – Conformance inspection - Continued.

4.4.1.6 Intermediate Assembly – Pallet Adapter PA125 Container, Unpainted (Drawing ACV0056) 1/	Conformance Criteria		Requirement Paragraph	Inspection Methods
	Qty	MIL-STD-1916 level		
Major 101 – Weld missing, incomplete or incorrect		VI	3.3	Visual
Major 102 – Overall length		III	3.3	Gage
Major 103 – Overall width		III	3.5	Gage
Major 105 – Material 2/			3.3	Visual
Minor 201 – Evidence of poor workmanship		II	3.8.1/3.8.3	Visual

NOTES:
1/ Next higher assembly Drawing is ACV0053.
2/ Per lot of material.

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TABLE IV – Conformance inspection - Continued.

4.4.1.7 Intermediate Assembly – Pallet Adapter PA125 Container, Painted (Drawing ACV0056) 2/	Conformance Criteria		Requirement Paragraph	Inspection Methods
	Qty	MIL-STD-1916 level		
Major 101 – Topcoat thickness (total thickness minus primer thickness (see AC200000423 and Note 2))		III	3.3	Gage
Major 102 – Marking missing, incorrect or illegible		III	3.3	Visual
Major 103 – Paint adhesion test	4		3.5	4.5.3
Major 104 – Primer coat thickness (see AC200000423 and Note 1)		III	3.3	Gage
Minor 201 – Evidence of poor workmanship		II	3.8.2	Visual

NOTES:

1/ The inspection is only applicable for CARC coating. The primer thickness may be determined from a test panel when representative of the part for CARC topcoat. When destructive testing is done on samples (e.g. ASTM-D4138) these surfaces may be touched up in accordance with 3.6.1 of MIL-DTL-53072

2/ Next higher assembly Drawing is ACV0053.

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TABLE IV – Conformance inspection - Continued.

4.4.1.8 Lifting Ring Retainer AC200000805, AC200000806 1/	Conformance Criteria		Requirement Paragraph	Inspection Methods
	Qty	MIL-STD-1916 level		
Major 101 – Legs proper height		III	3.3	Gage
Minor 201 – Evidence of poor workmanship		II	3.8	Visual
NOTES: 1/ Next higher assembly Drawings AC200000469, AC200000807, AC20000827, ACV00135, ACV00157, ACV00159, ACV00647, or ACV00648 applies, as applicable				

4.4.2 Inspection lot formation. The adapters shall be assembled into identifiable lots. Each lot shall consist of the same type of adapters, manufactured under essentially the same conditions. Component parts shall be inspected and tested. Production lot formation shall conform to requirements of MIL-STD-1168 and inspection sample formation shall conform to the requirements of MIL-STD-1916.

4.4.3 Lot rejection. The inspection acceptance criteria of accept on zero defect and reject on 1 defect applies to all inspection. The lot is rejected, not just the sample size.

4.5 Methods of inspection. (See 6.4)

4.5.1 Unit load test. (First article inspection only (see 6.2.d)). Unit loads for the painted pallet and adapter assemblies shall be fabricated in accordance with MIL-STD-1660 with the following exceptions. The gross weight of the ammunition load shall be 2,500 pounds minimum. Fabrication of the unit load shall include using the exact container for which the adapter is designed. The following tests (Table V) shall be performed on the unit loads in accordance with MIL-STD-1660. The tests shall be performed in the sequence indicated (“a” through “f”).

TABLE V – Unit load test

MIL-STD-1660	
Unit Load Test	Paragraph Section
a. Stacking test.	a. 5.4.1
b. Repetitive shock test.	b. 5.4.2 Appendix A
c. Edgewise drop test.	c. 5.4.3.1
d. Incline impact test.	d. 5.4.4 Appendix D
e. Sling compatibility test.	e. 5.4.8.1
f. Disassembly test.	f. 5.4.9

4.5.2 Lifting ring (bar) pull test. Lifting rings (bars) of the adapter top assembly shall be placed in a test machine designed for pull testing. The load shall be applied by a

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steady pressure and the load read directly by a gage. The load applied, in a vertical direction, to each ring shall be 2,500 pounds and shall be maintained for the required time. The results of the test shall not show signs of deformation, change in length of the ring (bar) nor any separation of the ring (bar) from top assembly.

4.5.3 Paint adhesion test. The paint adhesion shall be tested in accordance with paragraph 4.8.6 of TT-C-490.

5. PACKAGING

5.1 Packaging. For acquisition purposes, packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contracting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. These metal adapters are intended for use with metal pallets when unitizing packaged ammunition containers.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Type of adapter (see 1.2).
c. Applicable stock number.
d. First article sample requirements. Contractual provisions should require that the unit load test (as per MIL-STD-1660) is the responsibility of the contractor to accomplish with prior testing approval by the U.S. Army Development Command – Armaments Center (DEVCOM-AC), ATTN: FCDD-ACE-QME, Picatinny Arsenal, NJ 07086-5000
e. Packaging requirements, if other than specified in Section 5.
f. Certificate of Conformance for each lot of shipment.
g. Certificate of Conformance of conventional coating for TT-C-490 and Mil-STD-171.

6.3 Submission of contactor equipment designs. Copies of designs should be submitted to: U.S. Army Development Command – Armaments Center (DEVCOM-AC), ATTN: FCDD-ACE-QME, Picatinny Arsenal, NJ 07806-5000. This address should be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

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6.4 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to U.S. Army Development Command – Armaments Center (DEVCOM-AC), ATTN: FCDD-ACE-QME, Picatinny Arsenal, NJ 07806-5000. This description should include but not be limited to the accuracy and precision of the method, test data demonstrating the accuracy and precision and drawings of any special equipment required.

6.5 Identification of changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to extent of the changes.

6.6 Subject term (key word) listing

Lifting ring
Container
Unit load test

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APPENDIX A

QUALITY ASSURANCE PROVISIONS FOR REFURBISHMENT

A1. SCOPE

A1.1 Scope. This appendix provides the Quality Assurance provisions for the refurbishment of metal pallet adapters.

A2. APPLICABLE DOCUMENTS

A2.1 Government documents.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-1168	-	Ammunition Lot Numbering and Data card
MIL-STD-1916	-	Department of Defense Test Method Standard: DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil)

A2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

US ARMY DEFENSE AMMUNITION CENTER DRAWINGS

(see paragraph 2.2.2 of this specification)

A2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SAE INTERNATIONAL

SAE-AMS-STD-595	-	Colors used in Government Procurement

(Copies of these documents are available from https://www.sae.org/)

A 3. REQUIREMENTS:

A3.1 Cleaning. Parts shall be free of loose paint, dirt, grease, and rust.

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APPENDIX A

A3.2 Refinish. The parts shall be refinished in accordance with the applicable drawings with the following exception:

a. For refinishing of adapter AC200000807 (training rounds only), No. 5.1.1 or 5.2 shall be applied to all bare metal, followed by No. 20.1 to all surfaces in accordance with MIL-STD-171, color forest green No. 34079 per SAE-AMS-STD-595 may be used as an alternate.

b. An "R" shall be placed directly following the part number of refurbished adapters, using the same size stencil and color specified on the drawing.

A3.3 Welds. Welds shall be in accordance with ACV00515, Class 1.

A3.4 Stackability. The pallet shall not be deformed to the extent that it effects subsequent disintegration of the load stacking of pallets in storage.

A3.5 Lifting rings. The lifting rings shall not be damaged or deformed to the extent of preventing safe handling by a sling during top lifting operations.

A3.6 Adapters. The top lift shall not be deformed to the extent that it does not engage the containers.

A3.7 Workmanship. Workmanship shall be in accordance with 3.8. In addition, the pallets shall:

a. The pallet assembly shall be free of deformation or damage that affects stacking, slinging or assembly of the pallet.

b. The pallet assembly shall be free of damage or deformation to the pallet legs that does not allow four-way entry or prevents the pallet from resting evenly on a level surface.

c. The pallet assembly shall be free of damage to the bottom adapter if deformed to the extent that it does not mate with the pallet.

A4. VERIFICATION

A4.1 Examination. Examinations for major and minor characteristics shall be performed in accordance with Table A1 and MIL-STD-1916.

A4.2 Inspection lot formation. The adapters shall be assembled into identifiable lots. Each lot shall consist of the same type of adapters, manufactured under essentially the same conditions. Component parts shall be inspected and tested. Production lot formation shall conform to requirements of MIL-STD-1168 and inspection sample formation shall conform to the requirements of MIL-STD-1916.

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APPENDIX A

TABLE A1 – Quality Conformance inspection.

Metal pallet adapter (see drawings listed in 2.2.2).	Conformance Criteria		Requirement Paragraph	Inspection Methods
	Qty	MIL-STD-1916 level		
Major 101 – Welds missing or broken		VI	A3.3	Visual
Major 102 – Stacking surface deformed		VI	A3.4	Visual
Major 103 – Lifting rings damaged or deformed		VI	A3.5	Visual
Major 104 – Adapter damaged or deformed		VI	A3.6	Visual
Minor 201 – Marking, missing, incorrect or illegible		II	A3.7	Visual
Minor 202 – Protective finish damaged		II	A3.7	Visual
Minor 203 – Evidence of poor workmanship		II	A3.7	Visual

CONCLUDING MATERIAL

Custodian:
Army-AR

Preparing activity:
Army-AR
(Project 8140-2023-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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